PCR

M01 | ver 1.6 | Revised 2026-10-03

SUSTech-BIO 2026 iGEM Team

Purpose

Perform high-fidelity amplification of a target DNA fragment for gel analysis and molecular cloning using either Takara PrimeSTAR Max or ABclonal Gloria Nova HS 2X Master Mix V2. Both options use the shared 50 µL or 25 µL recipe below; select the matching template guidance and thermal program.

Materials & Equipment

Item	Manufacturer	Identifier / specification	Use / Specification
PrimeSTAR Max Premix (2×)	Takara Bio	R045A/B	Option A [1]
Gloria Nova HS 2X Master Mix V2	ABclonal	RK20731	Option B [2]
Forward and reverse primers	As supplied	Custom	Separate 10 µM working solutions
Purified template DNA	User sample	Not applicable	Known concentration; target sequence verified
Sterilized Type II water	Laboratory-prepared	Type II; sterilized	Primer dilution and reaction setup
PCR tubes, filter tips, pipettes	Axygen tips; laboratory equipment	For PCR	Accurate dispensing at 0.5-50 µL
Thermal cycler	Laboratory equipment	Programmable thermal block; heated lid	Run the selected enzyme program
Refrigerated centrifuge	Eppendorf	PCR-tube-compatible rotor	Brief spin-down
PCR-grade mineral oil (optional)	As available	PCR-grade evaporation barrier	Optional evaporation barrier; see Troubleshooting
Agarose gel system and DNA ladder	As available	For target size	Gel buffer, stain and loading dye; product check
Crushed ice	Laboratory ice maker	Crushed ice bath	Keep reagent tubes cold

Before You Start

Prepare separate 10 µM primer working solutions. For a 100 µM stock, mix 10 µL stock with 90 µL water. The reaction table assumes this working concentration.

Check the expected amplicon length and both primer sequences. A known amplifiable positive control may be included when establishing an assay.

Thaw and gently mix the chosen premix, then briefly spin. Assemble reactions on ice in a clean area separated from amplified DNA. Prepare separate mixes when comparing products.

Procedure

A. Prepare the shared reaction mixture

1. Select one master mix from Materials & Equipment. Calculate template volume V (µL) = required DNA mass (ng) / measured concentration (ng/µL), using the product-specific input table below. Dilute concentrated stocks to a reliably pipettable volume.

2. Prepare reactions with the shared recipe. For multiple samples using the same enzyme, primer pair and template volume, prepare a mixture without template, allowing about 10% extra for pipetting loss. Dispense it before adding templates.

Component	50 µL	25 µL	Final
Selected PCR master mix (2×)	25 µL	12.5 µL	1×
Forward primer (10 µM)	1 µL	0.5 µL	0.2 µM
Reverse primer (10 µM)	1 µL	0.5 µL	0.2 µM
Template DNA	V50 µL	V25 µL	See input table
Sterilized Type II water	(23 - V50) µL	(11.5 - V25) µL	To volume
Total	50 µL	25 µL	

Use separate 10 µM primer working solutions. V50 and V25 denote the template volume in each reaction size; recalculate water whenever template volume changes. The selected primer concentration is 0.2 µM. [1, 2]

Template input guide

Template	Takara: 50 µL	ABclonal: 50 µL
Plasmid DNA	0.01-1 ng	0.001-10 ng
Human genomic DNA	5-200 ng	10-100 ng
E. coli genomic DNA	0.1-200 ng	0.5-200 ng

These are product-specific starting ranges. For a 25 µL reaction, halve the listed DNA mass to preserve concentration; this is a proportional adaptation. This guide covers purified plasmid and genomic DNA. [1, 2]

3. Mix complete reactions gently and briefly spin. Select the actual reaction volume and enable the heated lid where available; otherwise use the mineral-oil overlay described in Troubleshooting.

B. Select the thermal program

4. Run the program matching the selected master mix. Use the same temperatures and times for either reaction volume; calculate extension time from the expected amplicon length.

Option A: Takara PrimeSTAR Max (R045A/B)

Stage	Temperature	Time	Repetitions
Denaturation	98 °C	10 s	30-35 cycles: denaturation, annealing, extension
Annealing	55 °C	5 or 15 s	Within cycle
Extension	72 °C	5 s/kb	Within cycle
Hold	4 °C	Until collection	After cycling

A separate initial denaturation or final extension is generally unnecessary. For primers ≤25 nt, use 5 s when both Tm values are ≥55 °C; otherwise use 15 s. Estimate Tm (°C) = 2(A + T) + 4(G + C) - 5 from nucleotide counts. For primers >25 nt, start at 5 s. [1]

Rapid extension guidance covers human genomic targets up to 6 kb and E. coli targets up to 10 kb. For longer targets, try 15-30 s/kb. [1] Above 200 ng template/50 µL (scaled: 100 ng/25 µL), reduce input or use 30-60 s/kb. [4]

Option B: ABclonal Gloria Nova HS 2X Master Mix V2 (RK20731)

Transfer chilled reactions to a cycler preheated to 95 °C. [2]

Stage	Temperature	Time	Repetitions
Initial denaturation	95 °C	3 min	Once
Denaturation	95 °C	15 s	25-35 cycles: denaturation, annealing, extension
Annealing	60 °C*	30 s	Within cycle
Extension	72 °C	10 s/kb	Within cycle
Final extension	72 °C	5 min	Once
Hold	4 °C	Until collection	After cycling

*Start at 60 °C. For primers >20 nt, use the lower primer Tm + 3 °C; for primers <20 nt, use the lower Tm. For exactly 20 nt, start at the default and optimize if needed. Complex targets or fragments >10 kb may need 15 s/kb. [2]

C. Inspect the product

5. Analyze a small aliquot alongside an appropriate DNA ladder and any positive control included. Select agarose concentration for the expected fragment size and add loading dye at its working dilution. Retain the remaining product for the chosen downstream workflow. See M02 for the gel check and interpretation.

p.s.

Label tubes with the master mix and program name. When changing enzyme, reload its complete program, including initial and final stages.

Use the stated 10 µM primer working solutions. Adding a 100 µM stock at the listed volumes gives a tenfold concentration error.

Recalculate water when template volume changes. For inconsistent repeats, check mixing, pipetting and seals; use a shared reaction mix.

Expected Outcome

A predominant product at the expected size supports successful amplification. Save the gel image with lane identities. Band size alone does not establish sequence identity.

Troubleshooting

For problems first seen on a gel, start with M02, then use PCR Optimization below if indicated.

Observed problem	Check first	Suggested action
Bubbles remain after brief centrifugation	Persistent surface bubbles or bubbles attached to the tube wall	Gently tap the closed tube to release wall-attached bubbles. For a persistent surface bubble, use a fresh nuclease-free pipette tip to gently touch and break it; if needed, aspirate only the bubble while withdrawing the tip, avoiding loss of reaction liquid. Use a new tip for each tube, recap and briefly spin down. Avoid blowing air into the mixture during pipetting. [6]
Reaction volume decreases after cycling	Heated-lid activation, temperature and contact; tube-cap seal	Briefly spin down to collect condensation. If volume remains reduced, check that the heated lid is enabled, reaches its specified temperature and contacts properly sealed tubes. If evaporation persists or the cycler has no heated lid, overlay the next reaction with PCR-grade mineral oil before cycling, using enough to cover the entire liquid surface and an amount compatible with the tube/cycler. Keep oil amounts consistent between tubes. [7]

PCR Optimization

Use these options after M02 has checked sample loading, electrophoresis and detection. A gel pattern suggests which check to make; it does not identify a cause by itself. Keep the selected enzyme-specific recipe and program as the starting point, and change one factor at a time.

O1. Low or absent amplification

Check primer identity, sequences, orientation and binding to the actual template. Confirm the 10 µM working dilution. If primer stock quality is suspect, reconstitute an unused dry aliquot and prepare a fresh working solution. Review template integrity, purity and input against the selected product guidance. [1, 2, 4]

Recheck the full matching thermal program, especially annealing and length-dependent extension. Use an annealing-temperature gradient within the selected enzyme guidance if needed. For ABclonal difficult or long targets, apply its extension guidance in Section B; do not transfer settings between enzymes. A dilution comparison can help assess suspected template inhibition, but also reduces template input. [1-4]

If a distinct, correctly sized product is present but more DNA is needed, run additional parallel PCRs and pool matching products for purification. This means additional reactions, not an automatic increase in cycle number. A faint, well-separated target can be recovered using M02; quantify the eluate to decide whether it meets the downstream requirement.

O2. Nonspecific amplification or short by-products

After M02 excludes poor separation or loading distortion, review the expected product map and primer specificity. Extra bands or a short band near the front are candidates for nonspecific amplification or primer dimers, respectively; position alone does not establish their identity.

Confirm primer working dilution and template input before testing annealing conditions. Follow the selected enzyme guidance: for PrimeSTAR Max, retain the specified short annealing duration; for ABclonal, evaluate annealing temperature and excessive primer/template input or cycling when appropriate. Redesign primers if the binding sites are unsuitable. Recover the intended band in M02 only if it is clearly separated. [1-3]

O3. Residual plasmid template

Consider residual template separately from newly amplified by-products when the PCR started from a plasmid. Uncut plasmid conformations can migrate differently from linear DNA, so apparent size alone cannot identify a residual-template band.

For residual Dam-methylated plasmid template containing DpnI recognition sites, see the DpnI digestion chapter. DpnI does not generally remove unmethylated nonspecific PCR amplicons; use O2 for those products. Confirm template methylation and site suitability before selecting digestion. [5]

Reference

[1] Takara Bio. PrimeSTAR Max DNA Polymerase Product Manual, R045A, v202011Da (internal page headers), pp. 3-6. Accessed 2026-09-15. Open source

[2] ABclonal. Gloria Nova HS 2X Master Mix V2, RK20731, datasheet M19B05V1.1, pp. 1-2. Accessed 2026-09-15. Open source

[3] Takara Bio USA. When optimizing PCR conditions, which conditions are particularly important? PCR optimization guidance; updated 2023-09-28. Accessed 2026-09-15. Open source

[4] Takara Bio USA. If no amplification products are obtained, what parameters should be considered first when troubleshooting? Includes PrimeSTAR Max-specific guidance; updated 2023-09-28. Accessed 2026-09-15. Open source

[5] New England Biolabs. DpnI, R0176: recognition-site methylation requirement and Product Notes. Accessed 2026-09-15. Open source

[6] Citri A et al. Comprehensive qPCR profiling of gene expression in single neuronal cells. Nature Protocols 7, 118-127 (2012), step 14 and Table 1. Physical bubble-removal technique adapted from qPCR wells to PCR tubes; author article mirrored by Gene Quantification. Accessed 2026-09-16. Open source

[7] Bio-Rad. PTC-100 Thermal Cycler Operations Manual, Bulletin 07512, sections Sealing Sample Vessels, Sealing with Oil or Wax, and Sealing with the Hot Bonnet Lid, printed pp. 4-5 to 4-6. General evaporation-control guidance; not an identification of the team cycler. Accessed 2026-09-16. Open source